

Minimal match and SNR loss from imperfect templates

J. Sfeir

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Abstract

We quantify the loss caused by an imperfect gravitational-wave template and the bank density required to control it in a deliberately restricted setting: one detector, stationary Gaussian noise, and a two-parameter non-spinning TaylorF2 bank. Owen’s matched-filter metric supplies the formal basis; a generation-wise, local-metric growth scheme is inspired by Cokelaer, but replaces his explicit connector graph with distance-based collision suppression and adds a greedy boundary repair. It is therefore not a faithful implementation of Cokelaer’s placement algorithm. At nominal $\text{MM} = 0.9700$ the population-averaged discretisation loss is well below the exact worst-case loss $1 - \text{MM}^3$, whereas waveform-family error dominates at high mass. The hexagonal-to-square template-count ratio changes with region and boundary treatment and cannot be treated as a universal lattice constant. Finally, an effective-trials fit suggests sublinear growth within each bank family, but it is calibrated far below the detection threshold and varies across the fitted tail. The resulting effective-volume curves are therefore conditional on that trials model and do not locate a robust optimum minimal match.

1 Scope and definitions

This is a controlled bank study, not a reproduction of a search. We use the `aLIGOZeroDetHighPower` power spectral density between 20 and 1024 Hz, non-spinning TaylorF2 waveforms terminated at f_{ISCO} , and component masses between 5 and $50 M_\odot$ with $m_1 \geq m_2$. Results are separated into a main region $M = m_1 + m_2 \leq 35 M_\odot$ and an extended region $35 < M \leq 100 M_\odot$. The latter exposes the failure of an inspiral-only model; it is not a recommended search domain. Real searches additionally require spin dimensions, multi-detector coincidence and rejection of non-Gaussian transients [4, 5, 6].

For normalized waveforms the match and mismatch are

$$\mathcal{M}(\lambda, \lambda + \Delta\lambda) = \max_{t_c, \phi_c} \langle \hat{h}(\lambda) | \hat{h}(\lambda + \Delta\lambda) \rangle, \quad \mu = 1 - \mathcal{M}. \quad (1)$$

In noise-free data the recovered SNR fraction is exactly $\mathcal{M}$. A nominal minimal match MM bounds the design mismatch by $1 - \text{MM}$ only when the bank truly covers the region. For sources uniform in Euclidean volume, the exact nominal worst-case volume loss is $1 - \text{MM}^3 = 0.08733$ at $\text{MM} = 0.97$; the population averages below instead use injections uniform in component mass and are not an astrophysical population model. Owen introduced minimal match as an experimental design choice [1]; modern searches often apply it as a percentile criterion rather than a strict worst-case guarantee [9].

2 Method: Owen baseline and our placement

Metric. We follow Owen [1, 2] and expand the mismatch locally,

$$\mu \simeq g_{ij} \Delta\lambda^i \Delta\lambda^j, \quad g_{ij} = \frac{1}{2} \langle \partial_i \psi_\perp | \partial_j \psi_\perp \rangle_w, \quad (2)$$

where the phase derivatives are projected orthogonally to constant phase and time, and the weight is proportional to $|h(f)|^2 / S_n(f)$. The coordinates are the chirp times (τ_0, τ_3) at $f_0 = 20$ Hz. We use the analytic non-spinning 3.5PN TaylorF2 phase metric, with PyCBC supplying the waveform and PSD, and keep the frequency support fixed at the base waveform’s TaylorF2 support when differentiating. A separate finite-difference calculation tests the quadratic predictor against a pre-registered relative-error limit of 0.10000. It passes at mismatch 0.03 but not at 0.05; only those discrete radii were tested, and conclusions below $\text{MM} = 0.97$ are therefore metric extrapolations.

The symmetric mass ratio is

$$\eta = \frac{m_1 m_2}{(m_1 + m_2)^2} \leq \frac{1}{4}. \quad (3)$$

The bound follows directly from $(m_1 - m_2)^2 \geq 0$, with equality only for equal masses. Thus $\eta = 1/4$ is the physical equal-mass boundary, not a fitted parameter. We reserve μ for mismatch.

Placement and relation to Cokelaer. Owen provides the metric geometry and the spacing associated with a target mismatch, but not the complete finite-region growth rule used here. We therefore grow square and hexagonal banks generation by generation, recomputing the local metric at each parent, in the spirit of Cokelaer’s two-dimensional inspiral placement [3]. We also retain his post-growth projection of cells that cross the physical $\eta = 1/4$ boundary; projecting during reproduction can collapse a two-dimensional neighbourhood onto that line and leave gaps along it.

There are two explicit departures. First, we did not implement Cokelaer’s mother–daughter and adjacent–daughter connectors. We chose one lattice-independent duplicate rule instead: a proposal is rejected when it lies within one half of the local lattice spacing of an existing template, measured with the parent’s metric. This keeps the same collision mechanism for both lattices, but it does *not* preserve the connectivity bookkeeping of Cokelaer’s algorithm and has not been shown equivalent to it. Second, after growth we greedily repair a strip within one covering radius of the region boundary, because the uncorrected banks left boundary holes. Consequently, “lattice only” and repaired template counts are properties of our implementation, not counts from a reproduction of Cokelaer. This distinction is essential when interpreting Fig. 1.

Covering is first sampled with injections uniform in component mass and then challenged by an adaptive search. Population volume loss is $1 - \langle \mathcal{M}^3 \rangle$. Thresholds are evaluated both with the naive number of template–time samples and with an equivalent independent-trials count fitted from simulated Gaussian-noise maxima.

3 Results

3.1 Waveform and discretisation losses

In the declared interpolation model, the TaylorF2 family plus bank loses 0.19320 of the volume in the main region, compared with 0.17580 for the interpolated family-only estimate. In the extended region these become 0.51486 and 0.50779. The exact continuous-family loss would bound the discrete-bank loss from below, but the 28-point interpolation has no demonstrated one-sided error. Within this model, waveform-family error dominates discretisation in the extended region.

At $MM = 0.97$, the four banks give:

	main hex.	main square	ext. hex.	ext. square
templates	1772	2273	336	488
sampled covering fraction	1.00000	1.00000	0.99933	1.00000
population volume loss	0.02088	0.01534	0.01285	0.00945

Table 1: Measured banks at nominal $MM = 0.97$. Covering fractions use uniform injections and are not certificates of strict covering.

Uniform sampling misses small holes. Adaptive optimisation reaches mismatch 0.0322 in the main hexagonal bank and 0.0362 in the extended hexagonal bank, both above the 0.03 target; the two square banks survive the same search. These values are witnesses that refute strict covering, not certified global maxima. This is also why the population averages in Table 1 cannot be read as worst-case bounds.

The measured ratios at $MM = 0.97$ are 0.7796 (main) and 0.6885 (extended). Their variation with region, minimal match and boundary treatment shows that the measured count ratio is not a universal property of square and hexagonal lattices. Boundary sensitivity is expected: 0.79900 of main-region injections and 0.93833 of extended-region injections lie within one covering radius of an edge.

The population-averaged discretisation losses are substantially smaller than the exact nominal worst case (Fig. 2), because typical injections do not fall at a cell’s worst point. This statement concerns the declared uniform component-mass measure; another source distribution would produce another average.

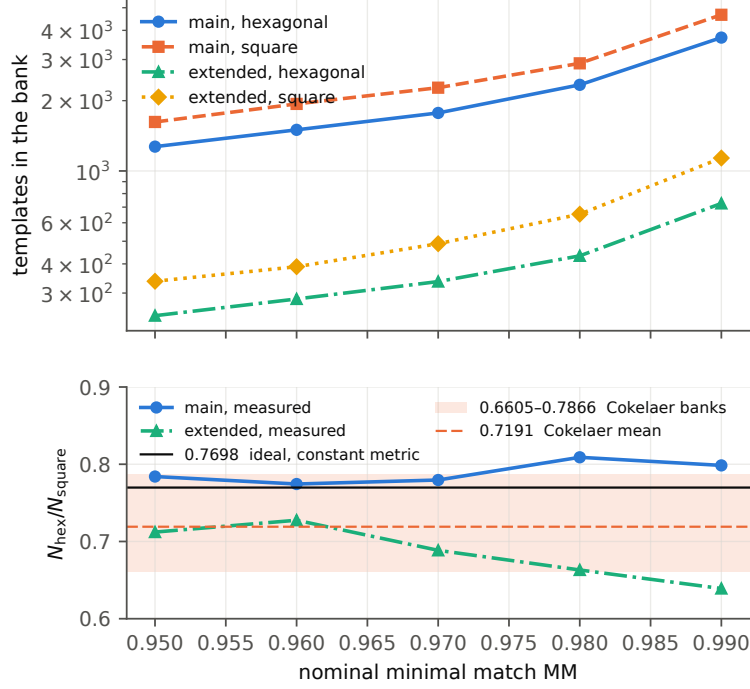


Figure 1: Bank size and hexagonal-to-square count ratio. The constant-metric ideal is 0.7698. The Cokelaer band is the full 0.6605–0.7866 range recomputed from the banks in his Tables I–II; 0.7191 is their mean, not a narrow uncertainty interval. Our banks include a different collision rule and boundary repair, so agreement or disagreement is descriptive rather than a validation of his algorithm. This claim would fail if both regions followed one MM-independent ratio near the lattice ideal.

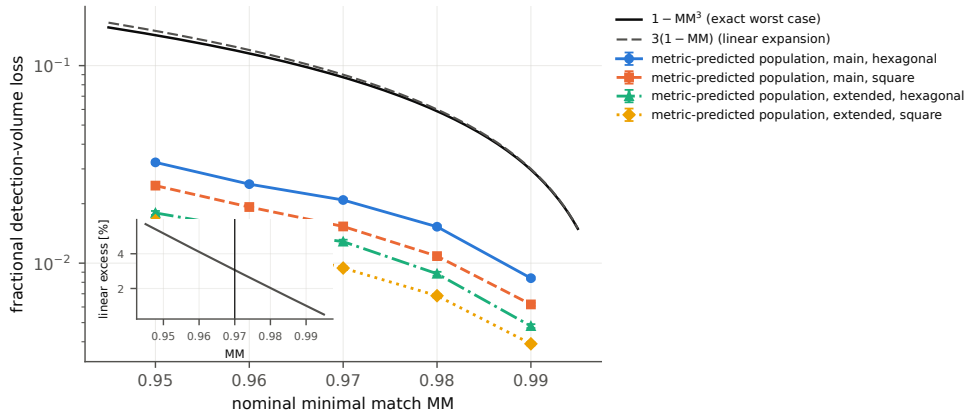


Figure 2: Exact nominal worst-case loss, its linear approximation, and metric-predicted population-average discretisation loss; the inset resolves the analytic-curve difference. Points below $MM = 0.97$ lie outside the validated quadratic range. The worst-case reading would fail if the bootstrap-resolved population points lay on its curve.

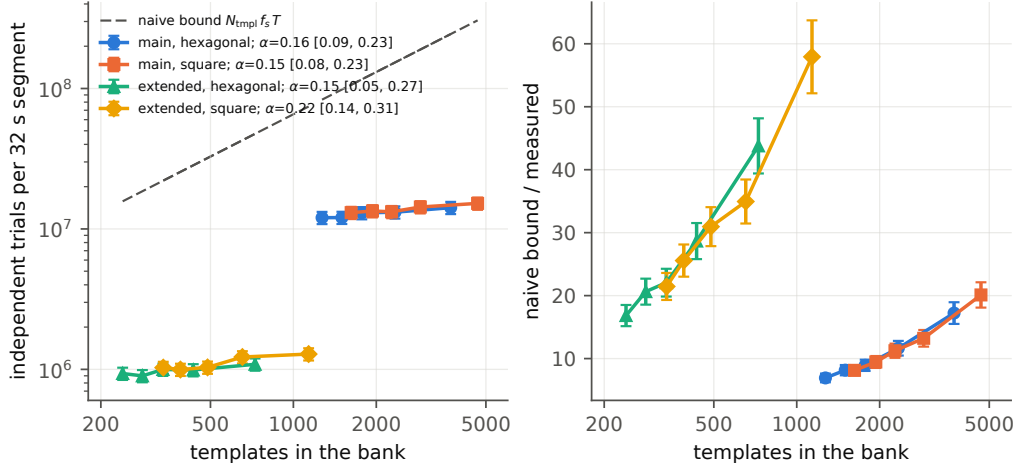


Figure 3: Equivalent independent trials inferred from Gaussian-noise maxima versus the naive template–time count; error bars and legend intervals expose fit precision. Independent templates would put the points on the dashed line, while linear growth would make each family parallel to it; neither occurs. The fitted count remains tail-dependent.

3.2 Threshold and effective volume

For independent complex-Gaussian trials,

$$\text{FAP} = 1 - (1 - e^{-\rho^{*2}/2})^N. \quad (4)$$

Counting every template and time sample as independent gives a conservative upper bound on the FAP for the finite correlated Gaussian grid [10], but it is loose. Fitting the maxima obtained by filtering simulated noise gives $\nu_{\text{eff}} = 4.063 \times 10^5$ equivalent independent trials per second for the main hexagonal MM = 0.97 bank. Within each fixed region and lattice, fits of $N_{\text{eff}} \propto N_{\text{templates}}^\alpha$ give $\alpha = 0.1568, 0.1500, 0.1516$ and 0.2214 . Pooling unlike families reverses the result ($\alpha = 1.2922$), demonstrating that bank size alone does not define the trials factor.

The calibration is made where simulated maxima occur and extrapolated to a substantially higher detection threshold. Moreover, threshold-wise estimates of ν_{eff} vary across the fitted tail. The fit is therefore a useful empirical model, not a measured false-alarm rate at the operating point.

Define a relative sensitivity proxy

$$V_{\text{eff}} = \frac{\langle \mathcal{M}^3 \rangle}{\rho^{*3}}. \quad (5)$$

Across the sampled minimal-match grid, the main hexagonal value changes by -1.87% under naive counting but by 1.76% under the fitted trials model. Thus even the sign depends on how trials are modelled. The calibrated curves are nearly flat and do not resolve a robust optimum; computational cost, deliberately absent from this proxy, would also enter a practical optimisation [7, 8].

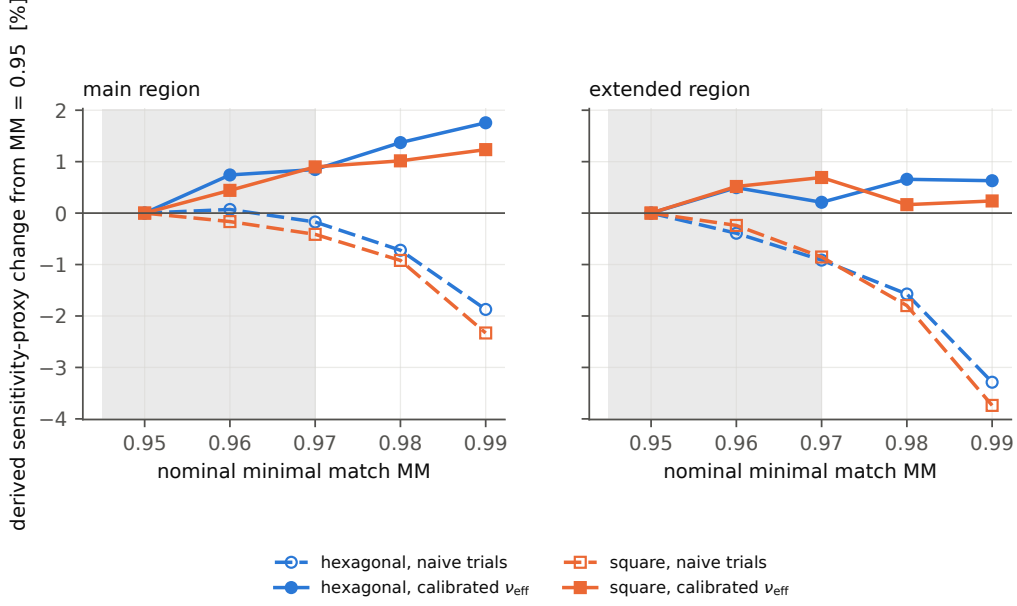


Figure 4: Derived relative sensitivity proxy under naive and fitted trials scenarios. No uncertainty bars are shown because the dominant ν_{eff} and tail-extrapolation uncertainty is unquantified. Agreement of the two scenarios would refute the claimed model sensitivity; instead their trends differ, and the curves do not establish an optimum minimal match.

4 Limitations and conclusion

The delivered banks have not been rechecked template by template with waveform overlaps; their covering results use the quadratic metric. The planned globally anchored metric, discard-versus-project placement sensitivity, Cokelaer connector implementation and SVD cross-check were not completed. The IMRPhenomD comparison composes continuous-family and discretisation effects approximately. Most importantly, the effective-trials calibration does not reach the detection tail.

Within those limits, the robust conclusions are narrower than the initial hypothesis. Mismatch gives the noise-free fractional SNR loss exactly. In the banks actually built at $\text{MM} = 0.97$, adaptive searches found losses of at least 0.0322 (main) and 0.0362 (extended) for the hexagonal placement, above the nominal 0.03; both square banks survived the same search. The metric-predicted population-average volume losses are 0.02088 versus 0.01534 in main and 0.01285 versus 0.00945 in extended. Against IMRPhenomD, the approximate hexagonal-bank loss rises from 0.19320 to 0.51486; the corresponding continuous-family estimates, 0.17580 and 0.50779, show that this main–extended contrast is mostly waveform-family error rather than discretisation.

Hexagonal placement uses 1772 instead of 2273 templates in main and 336 instead of 488 in extended: count ratios 0.7796 and 0.6885. At fixed nominal minimal match, the delivered square banks trade a larger count for lower population-average discretisation loss and survived the adaptive covering search. The hexagonal banks use fewer templates, but the implementations tested here contain small covering holes. Densifying hexagonal placement reduces its loss, so these results do not establish that one lattice is universally better; the choice depends on whether the objective is count, average loss, strict coverage, or effective detection volume. With 0.79900 and 0.93833 of injections within one covering radius of an edge, finite-boundary treatment is material. The square result is not a covering proof, and no preferred minimal match is established because the small effective-volume trend changes with the trials model.

Reproducibility. Headline measured values referenced with `\dataref` are generated from `data/project_numbers.json`; figures are produced by `scripts/make_figures.py`. The repository also records the algorithmic departures described in Sec. 2.

References

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